The documentation and process conversion measures necessary to comply with this revision shall be completed by 11 May 05.

INCH-POUND

MIL-PRF-19500/688
11 February 2005

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTOR, N-CHANNEL,
SILICON TYPES 2N7002, 2N7002T2, AND 2N7002UB
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a N-channel, enhancement-mode, low-threshold logic level, high frequency, high switching speed MOSFET, transistor. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-18), figure 2 (TO-205AD), and figure 3 (UB).

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	P_T T_A = +25°C (1) (2)	P_T T_C = +25°C (1) (2)	P_T $T_{SP(IS)}$ = +25°C (1) (2)	$R_{\theta JA}$ (2) (3)	$R_{\theta JC}$ (2) (3)	$R_{\theta JSP(IS)}$ (2) (3)
	mW	W	W	°C/W	°C/W	°C/W
2N7002	500	1.0	N/A	325	150	N/A
2N7002T2	800	5	N/A	175	30	N/A
2N7002UB	500	N/A	1.5	325	N/A	90

Type	V_{DS}	V_{GS}	I_{D1} (4) T_C = +25°C	I_{D2} (4) T_C = +100°C	I_{DM}	T_J and T_{STG}
	V dc	V dc	A dc	A dc	A(pk)	°C
2N7002 2N7002T2 2N7002UB	60	± 40	0.2	0.073	.8	-65 to +200

(1) For derating, see figures 4, 5, and 6.
(2) See 3.7.
(3) For thermal curves, see figures 7, 8, 9, and 10.
(4) The following formula derives the maximum theoretical I_D limit. I_D is limited by package and internal wires and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)\ DSS}$ V_{GS} = 0 V I_D = 10 μA dc	$V_{GS\ (th)1}$ V_{DS} = V_{GS} I_D = 0.25 mA dc	Max I_{DSS1} V_{DS} = 60V V_{GS} = 0 V	Max $r_{DS(on)}$ (1) V_{GS} = 10 V dc	
			V_{DS} = 60 percent of rated V_{DS}	T_J = +25°C at I_{D1}	T_J = +125°C at I_{D2}
	V dc	V dc Min Max	μA dc	Ohm	Ohm
2N7002 2N7002T2 2N7002UB	60	1 2.5	0.1	4	13.5

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

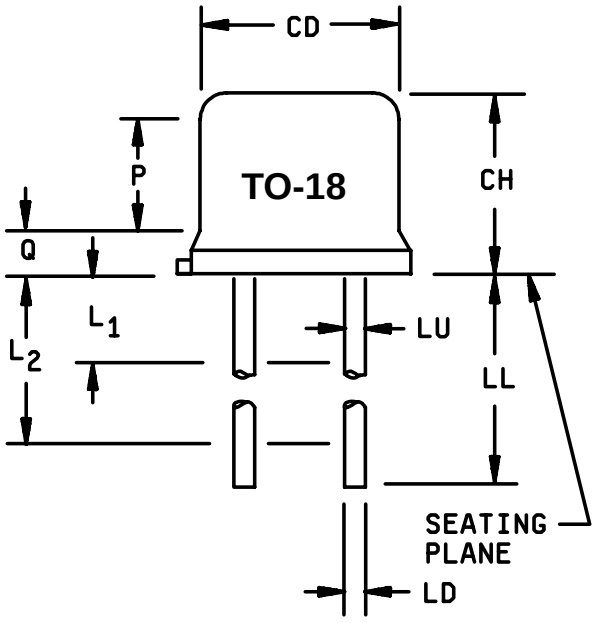
MIL-STD-750 - Test Methods for Semiconductor Devices.

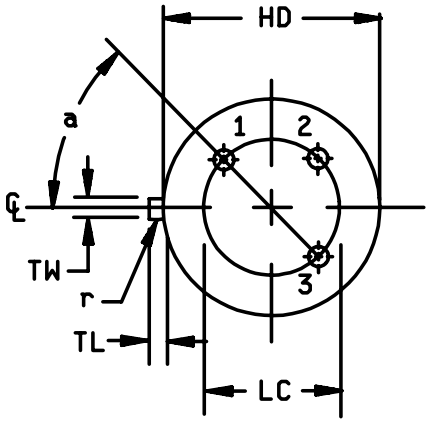
(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8
LU	.016	.019	0.41	0.48	7,8
L1		.050		1.27	7,8
L2	.250		6.35		7,8
P	.100		2.54		
Q		.030		0.76	5
TL	.028	.048	0.71	1.22	3,4
TW	.036	.046	0.91	1.17	3
r		.010		0.25	10
α	45° TP		45° TP		6
1, 2, 9, 11, 12					





NOTES:
1. Dimension are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 2.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The drain shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 = source, lead 2 = gate, lead 3 = drain.

FIGURE 1. Physical dimensions, 2N7002, TO-206AA (formerly TO-18)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

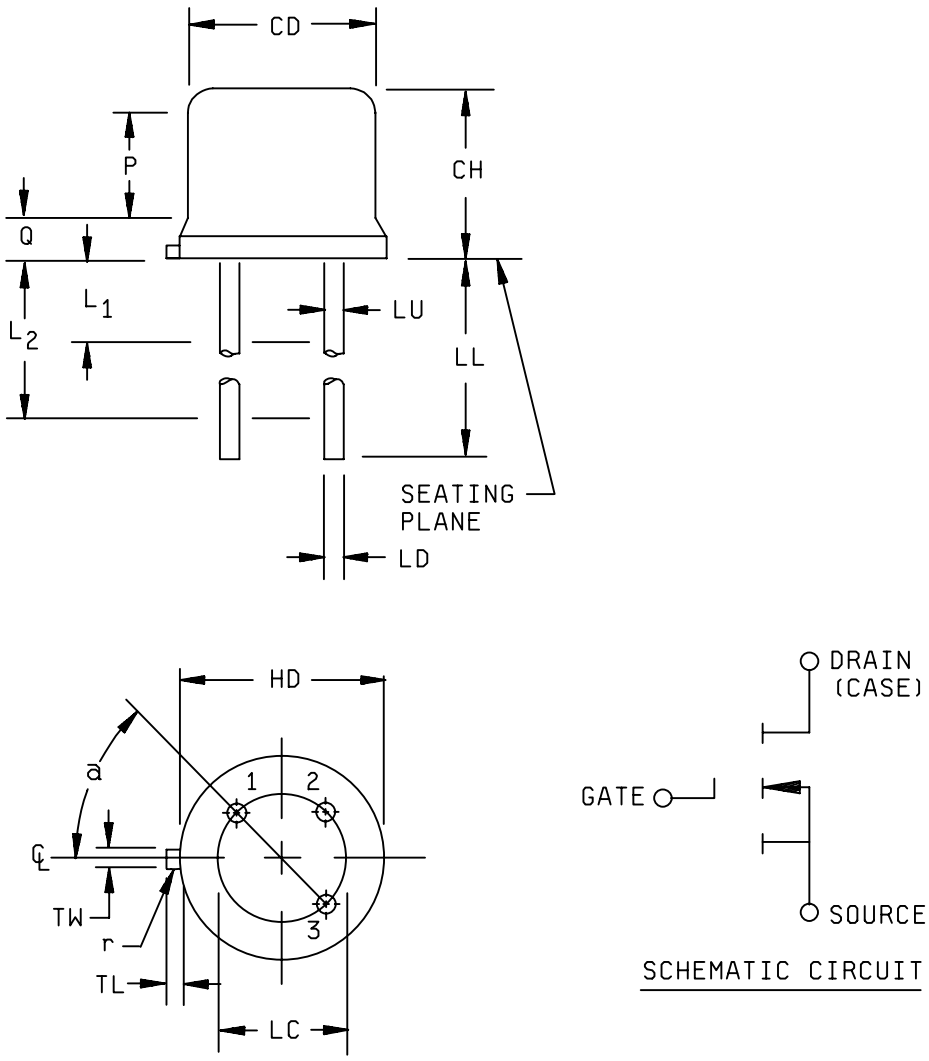


FIGURE 2. Physical dimensions 2N7002T2, (TO-205AD).

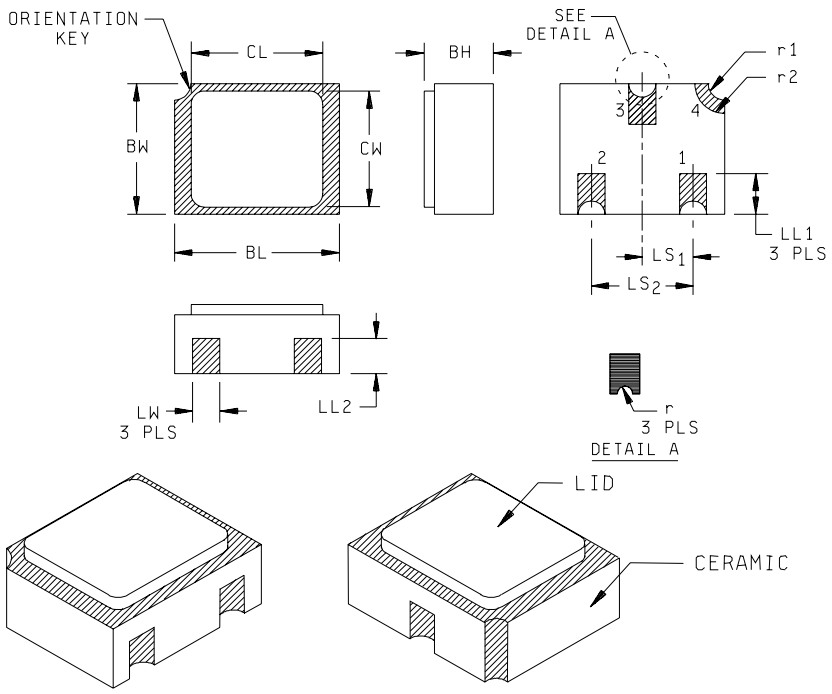
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
TW	.028	.034	0.71	0.86	2
TL	.029	.045	0.74	1.14	3
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8
LC	.200 TP		5.08 TP		6
LU	.016	.019	0.41	0.48	7,8
L_1		.050		1.27	7,8
L_2	.250		6.35		7,8
P	.100		2.54		5
Q		.050		1.27	4
r		.010		0.25	9
α	45 TP		45 TP		6

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Beyond radius (r) maximum, TW shall be held for a minimum length of .011 (0.028 mm).
3. Dimension TL measured from maximum HD.
4. Outline in this zone is not controlled.
5. Dimension CD shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 + .001, - .000 (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 (0.18 mm) radius of true position (TP) at MMC relative to tab at MMC.
7. LU applies between L_1 and L_2. LD applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. Radius (r) applies to both inside corners of tab.
10. Drain is electrically connected to the case.
11. Lead 1 = source, lead 2 = gate, lead 3 = drain.
12. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions 2N7002T2, (TO-205AD) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.



Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	1,2,3,4
BH	.046	.056	0.97	1.42	
BL	.015	.128	2.82	3.25	
BW	.085	.108	2.41	2.74	
CL		.128		3.25	
CW		.108		2.74	
LL1	.022	.038	0.56	0.96	
LL2	.017	.035	0.43	0.89	
LS1	.036	.040	0.91	1.02	
LS2	.071	.079	1.81	2.01	
LW	.016	.024	0.41	0.61	
r		.008		0.20	
r1		.012		0.31	
r2		.022		0.56	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas. Lid on package is a metallized area.
4. Pad 1 = Gate, Pad 2 = Source, Pad 3 = Drain, Pad 4 = shielding connected to the lid.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions, 2N7002UB, surface mount (UB version).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1 (TO-205AD), figure 2, and figure 3 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.6.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III.).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and herein.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500) (1) (2)	Measurement	
	JANS level	JANTX and JANTXV levels
3a	Temperature cycling Test condition C.	Temperature cycling Test condition C.
(3) 3	Gate stress test (see 4.5.2)	Gate stress test (see 4.5.2)
(3) (4) 3c	Method 3161 of MIL-STD-750 (see 4.3.1)	Method 3161 of MIL-STD-750 (see 4.3.1)
9	I_{GSS1}, I_{DSS1},-subgroup 2 of table I herein	Not Applicable
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	Subgroup 2 of table I herein; I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, ΔI_{GSS1} = ± 10 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 1 μA dc or ±100 percent of initial value, whichever is greater.	Subgroup 2 of table I herein. I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$
12	Method 1042 of MIL-STD-750, test condition A. t = 240 hours	Method 1042 of MIL-STD-750, test condition A
13	Subgroups 2 and 3 of table I herein; ΔI_{GSS1} = ± 10 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 1 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value or ± 0.5 ohm, whichever is greater. $\Delta V_{GS(th)1}$ = ± 10 percent of initial value or ± 0.3 V dc.	Subgroup 2 of table I herein; ΔI_{GSS1} = ± 10 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 1 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value or ± 0.5 ohm, whichever is greater. $\Delta V_{GS(th)1}$ = ± 10 percent of initial value or ± 0.3 V dc.
14	Fine leak method 1071 of MIL-STD-750, test condition H 5 x 10^{-8} atm cc/s. Gross leak method 1071 of MIL-STD-750, test condition C.	Fine leak method 1071 of MIL-STD-750, test condition H 5 x 10^{-8} atm cc/s. Gross leak method 1071 of MIL-STD-750, test condition C.

(1) At the end of the test program, I_{GSSF1} and I_{GSSR1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1} and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime before screen 9.
(4) This test need not be performed in group A when performed as a screen.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

4.3.1 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). The $Z_{\theta JX}$ limit used in screen 3c of 4.3 shall comply with the thermal impedance graph in figures 7, 8, 9, and 10 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3161.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for quality conformance inspection in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa of MIL-PRF-19500 (JANS) and 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table III herein. See 4.4.2.2 herein and table VIb of MIL-PRF-19500 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) requirements shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table III herein.

4.4.2.1 Group B inspection table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1051	Test condition C.
B4	1042	Test condition D; 2,000 cycles. The heating cycle shall be 1 minute minimum.
B5	1042	Accelerated steady-state reverse bias power; test condition A, V_{DS} = rated T_A = +175°C minimum, t = 120 hours.
B5	1042	Accelerated steady-state gate stress; condition B, V_{GS} = rated, T_A = +175°C minimum, t = 24 hours.
B5	2037	Bond strength (Al-Au die interconnects only); test condition A.
B6	3161	See 4.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

4.4.2.2 Group B inspection, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1051	Test condition C, 25 cycles.
B3	1042	Test condition D.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table III herein.

Subgroup	Method	Condition
C2	2036	Test condition E, (not applicable for surface mount devices).
C5	3161	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3)
C6	1042	Test condition D.

4.4.4 Group E inspection. Group E inspection shall be performed for qualification or re-qualification only. Group E inspection shall be conducted in accordance with tests and conditions specified for subgroup testing in table IX of MIL-PRF-19500 and table II herein. Electrical measurements shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Gate stress test.

a. V_{GS} = 24 V minimum.

b. t = 250 μs minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance	3161	See 4.3.1	$Z_{\theta JC}$			°C/W
Breakdown voltage, drain to source	3407	Bias condition C, V_{GS} = 0 V; I_D = 10.0 μA dc	$V_{(BR)DSS}$	60		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = 250 μA dc	$V_{GS(th)1}$	1.0	2.5	V dc
Gate current	3411	Bias condition C; V_{DS} = 0 V; V_{GS} = +15 and -15 V dc	I_{GSS1}		±10	nA dc
Drain current	3413	V_{GS} = 0; bias condition C; V_{DS} = 60V	I_{DSS1}		100	nA dc
Drain to source on-state voltage	3405	V_{GS} = 10 V dc; condition A; pulsed (see 4.5.1), I_D = 0.8 A dc	$V_{DS(on)1}$		7.5	V
Drain to source on-state voltage	3405	V_{GS} = 4.5 V dc; condition A; pulsed (see 4.5.1), I_D = 0.3 A dc	$V_{DS(on)2}$		10	V
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc; condition A; pulsed (see 4.5.1); I_D = 0.250 A dc	$r_{DS(on)1}$		4.0	ohms
Static drain to source on-state resistance	3421	V_{GS} = 5 V dc; condition A; pulsed (see 4.5.1); I_D = 0.200 A dc	$r_{DS(on)2}$		6.0	ohms
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1); V_{GS} = 0 V I_S = 0.2 A dc	V_{SD}	0.7	1.2	V (pk)
Forward transconductance	3475	Pulsed (see 4.5.1), $V_{DS} = V_{GS}$ = 10 V dc I_D = 0.2 A	g_{FS}	80		mS
Subgroup 3						
High temperature operation:		T_C = +125°C				
Gate current	3411	Bias condition C, V_{DS} = 0 V; V_{GS} = +20 V dc and -20 V dc	I_{GSS2}		± 500	nA dc

See footnote at the end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued.						
Drain current	3413	Bias condition C, V_{GS} = 0 V, V_{DS} = 60V	I_{DSS2}		10	µA dc
Drain to source on-state voltage	3405	V_{GS} = 10 V dc; condition A; pulsed (see 4.5.1), I_D = 0.5 A dc	$V_{DS(on)3}$		6.5	V
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc; condition A; pulsed (see 4.5.1); I_D = 0.5 A dc	$r_{DS(on)3}$		13.5	ohms
Gate to source voltage (threshold)	3403	V_{DS} = V_{GS}, I_D = 250 mA dc	$V_{GS(th)2}$	1.0	2.5	V dc
Low temperature operation:		T_C = -55°C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = 1.0 mA	$V_{GS(th)3}$		2.5	V dc
Subgroup 4						
Switching time test	3472	R_{gen} = 50 Ω; R_{GS} = 50 Ω,				
Turn-on delay time		V_{DD} = 30 V dc; $I_D \cong$ 2 A dc	$t_{d(on)}$		20	ns
Turn-off delay time		V_{DD} = 30 V dc; $I_D \cong$ 2 A dc	$t_{d(off)}$		20	ns
Small signal common source short-circuit input capacitance	3431	V_{DS} = 25 V dc; V_{GS}; = 0 V; f = 1 MHz	C_{iss}		50	pF
Small signal common source short-circuit output capacitance		V_{DS} = 25 V dc; V_{GS}; = 0V; f = 1 MHz,	C_{oss}		25	pF
Small signal common source short-circuit reverse transfer capacitance	3433	V_{DS} = 25 V dc; V_{GS}; = 0 V; f = 1 MHz	C_{rss}		5	pF
Subgroup 5						
Safe operating area test	3474	V_{DS} = 80 percent of rated V_{DS} (see 1.3)				
Electrical measurements		See table I, subgroup 2.				

See footnote at the end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroups 6 and 7 Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Electrical measurements		See table I, subgroup 2. see table III herein.	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2 1/			45 devices c = 0
Steady-state reverse bias	1042	Condition A; 1,000 hours	
Electrical measurements		See table I subgroup 2.	
Steady-state reverse bias	1042	Condition B; 1,000 hours	
Electrical measurements		See table I subgroup 2.	
Subgroup 4			Sample size N/A
Thermal impedance curves		Each supplier shall submit their qual-lot average and design thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and $Z_{\theta JX}$ limit shall be provided to the qualifying activity in the qualification report.	
Subgroup 5			
Not applicable			
Subgroup 6			3 devices
ESD	1020	Not required for devices classified as ESD class 1	
Subgroup 7			22 devices, c = 0
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476		

1/ A separate sample may be pulled for each test condition.

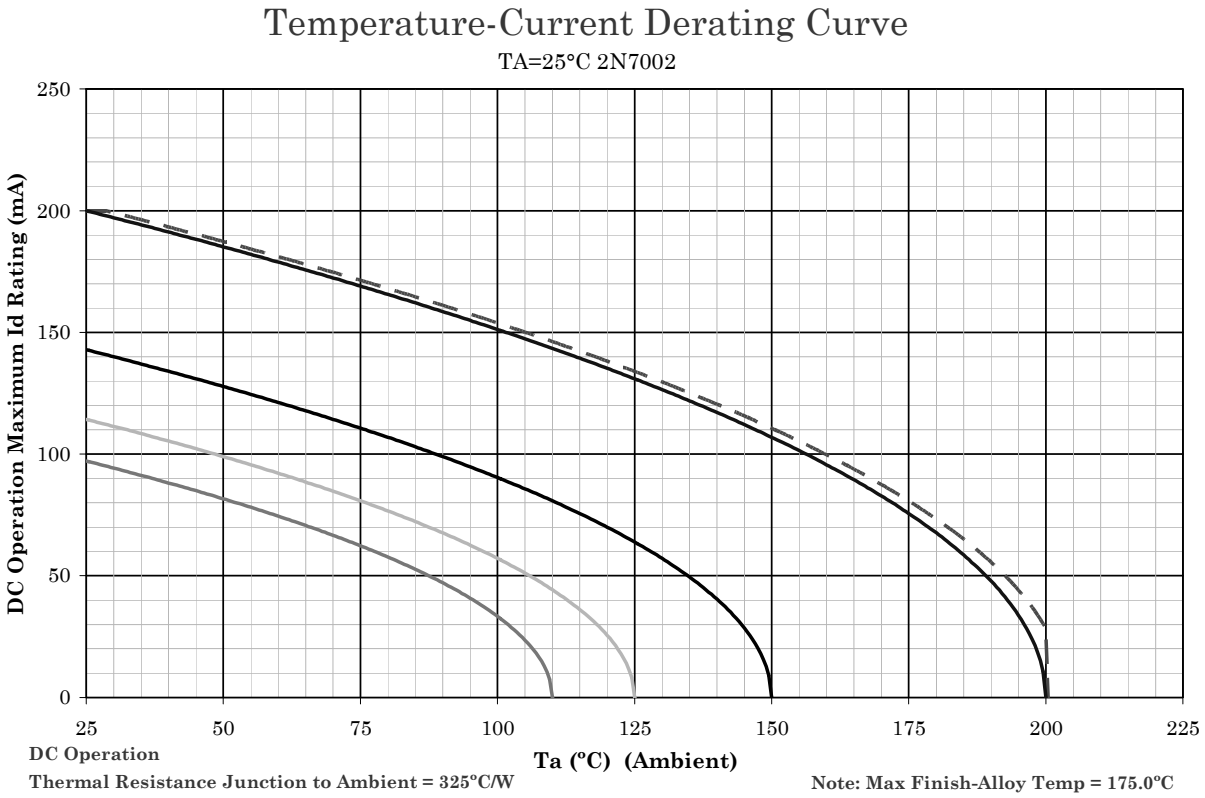
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

TABLE III. Group B, C, and E delta measurements. 1/ 2/ 3/ 4/

Step	Inspection 5/	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Gate current	3411	Bias condition C; V_{DS} = 0 V V_{GS} = +15 V dc and -15 V dc	ΔI_{GSS}	± 10 nA dc or 100 percent of initial value, whichever is greater.		
2.	Drain cut off current	3413	Bias condition C; V_{DS} = +60 V V_{GS} = 0 V dc	ΔI_{DSS}	± 1 μA dc or 100 percent of initial value, whichever is greater.		
3.	Small signal, drain to source on-state resistance	3423	V_{GS} = 5 V dc; condition A, pulsed (see 4.5.1); I_D = .2 mA dc	$\Delta r_{DS(on)}$	± 0.5 ohm or ± 20 percent of initial value, whichever is greater.		
4.	Gate to source voltage (threshold)	3403	V_{DS} = V_{GS} ; I_D = 0.25 mA dc	$\Delta V_{GS(th)}$	± 10 percent of initial value or ± 0.3 V dc, whichever is greater.		
5.	Thermal impedance	3161	See 4.3.1	$\Delta Z_{\theta JX}$	± 10 percent of initial value, see figures 7, 8, 9, and 10		

1/ The delta measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table III herein, step 1, 2, 3, 4, and 5.
   b. Subgroup 4, see table III herein, step 1, 2, 3, 4, and 5.
2/ The delta measurements for table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500 are as follows: Subgroup 3, see table III herein, step 1, 2, 3, 4, and 5.
3/ The delta measurements for table VII of MIL-PRF-19500 are as follows: Subgroup 6, see table III herein, step 5.
4/ The delta measurements for table IX of MIL-PRF-19500 are as follows: Subgroup 1, see table III herein, step 1, 2, 3, 4, and 5.
5/ See MIL-PRF-19500 for sampling plan.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

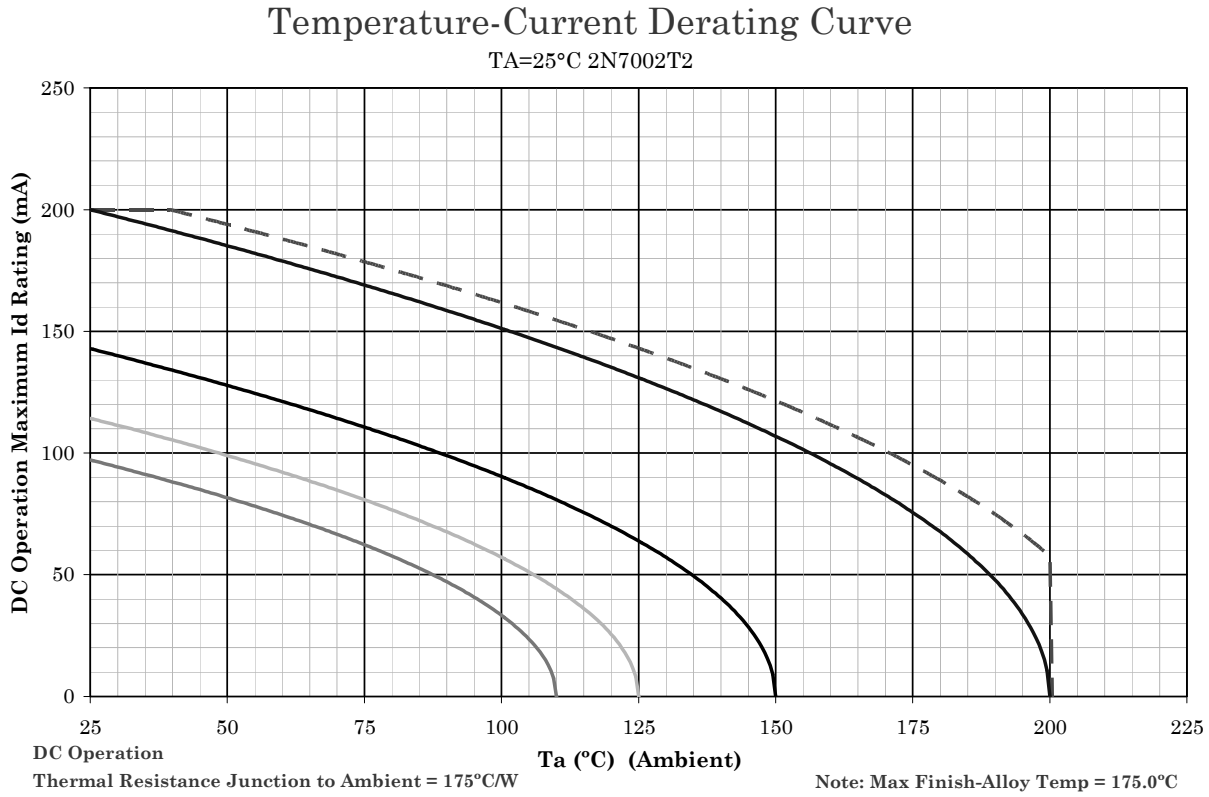


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 4. Derating for 2N7002 ($R_{\theta JA}$) TO-18.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

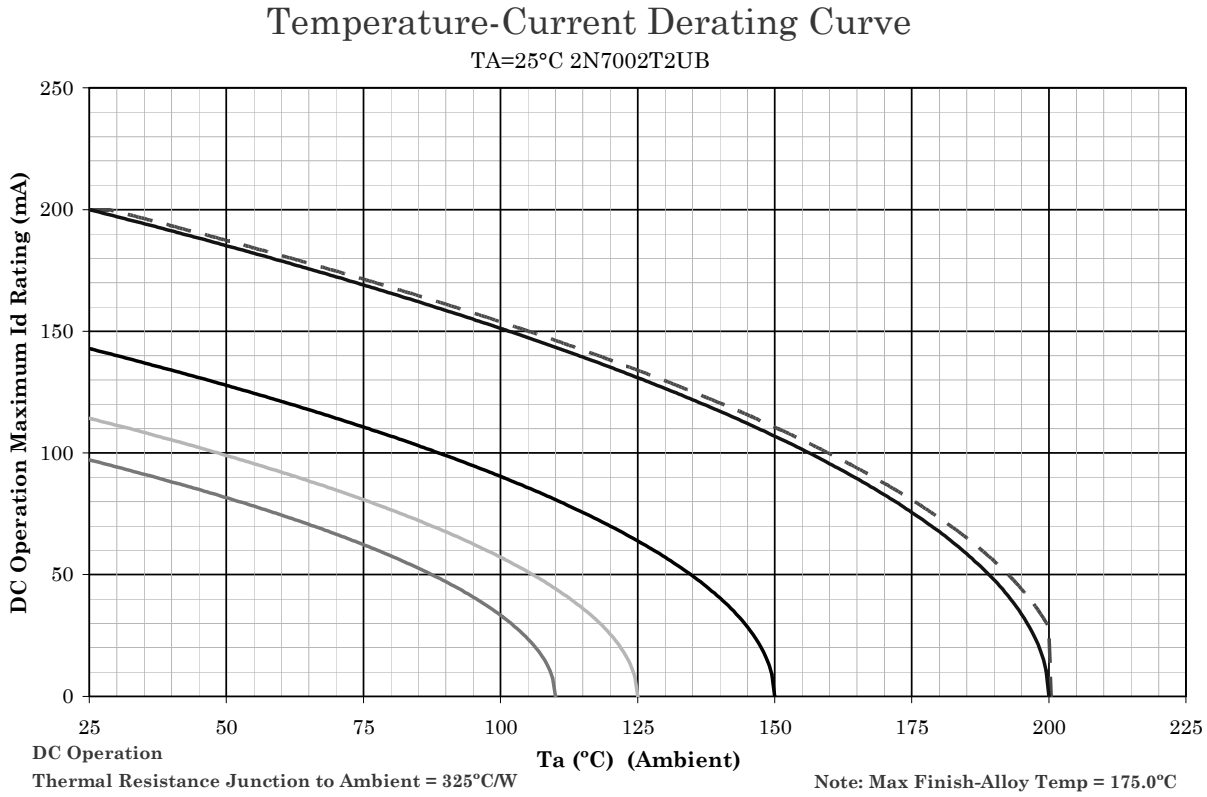


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 5. Derating for 2N7002T2 ($R_{\theta JA}$) TO-39.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.



NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 6. Derating for 2N7002UB ($R_{\theta JA}$).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

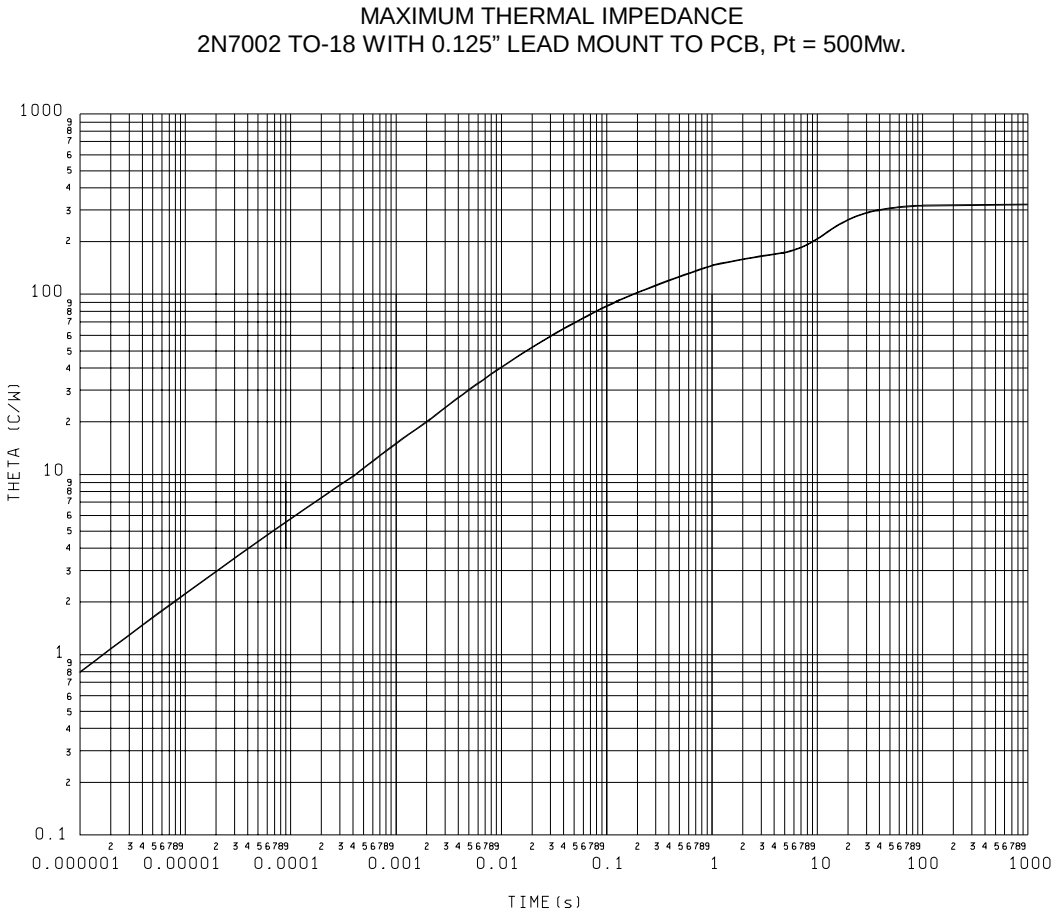


FIGURE 7. Thermal impedance graph ($R_{\theta JA}$) for 2N7002 (TO-18).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

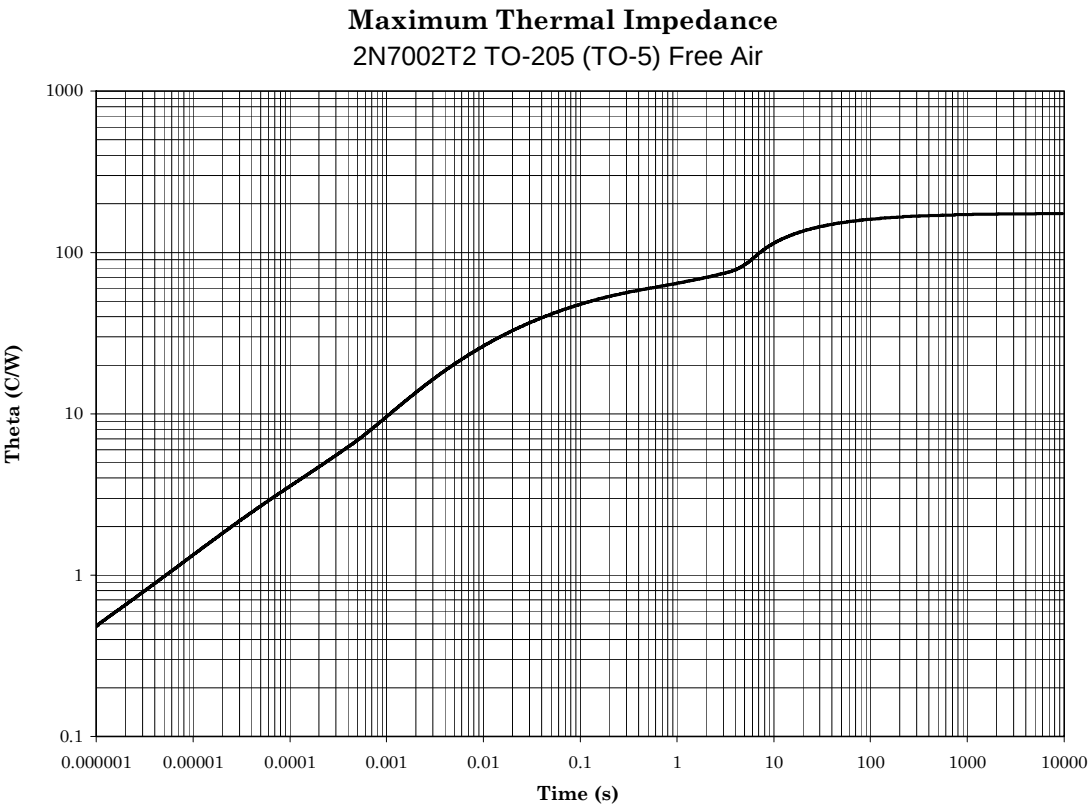


FIGURE 8. Thermal impedance graph ($R_{\theta JA}$) for 2N7002T2 (TO-205AD).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

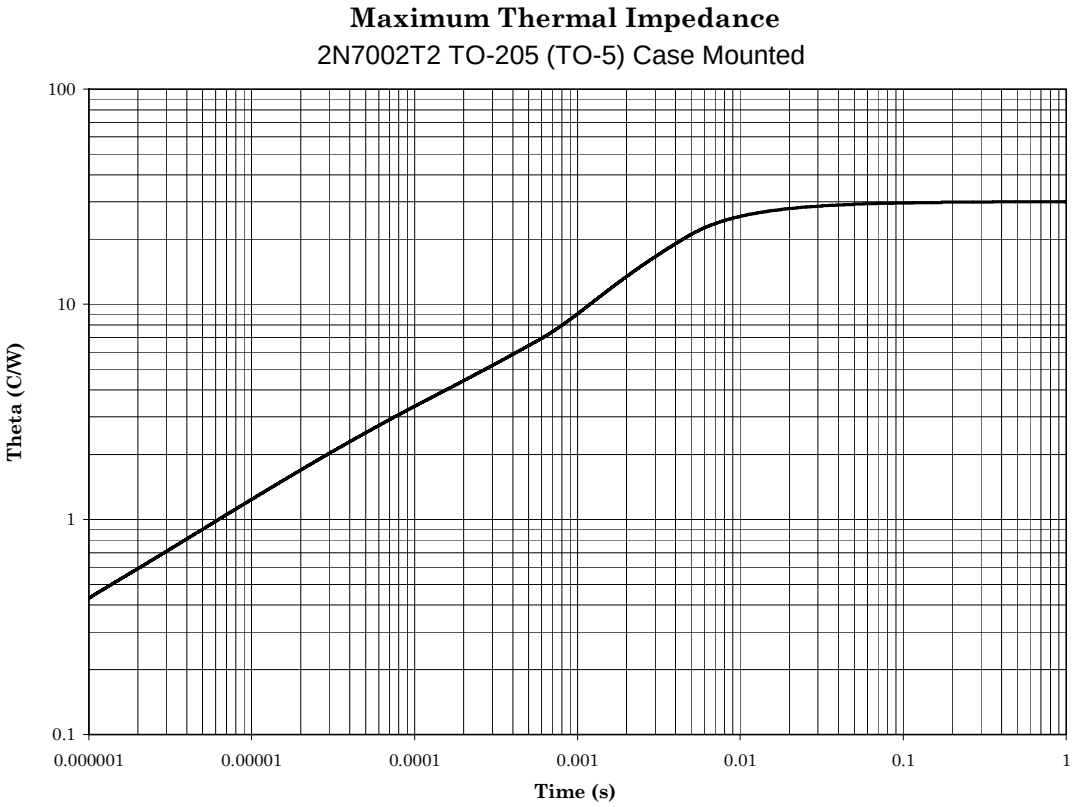


FIGURE 9. Thermal impedance graph ($R_{\theta JC}$) for 2N7002T2 (TO-205AD).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

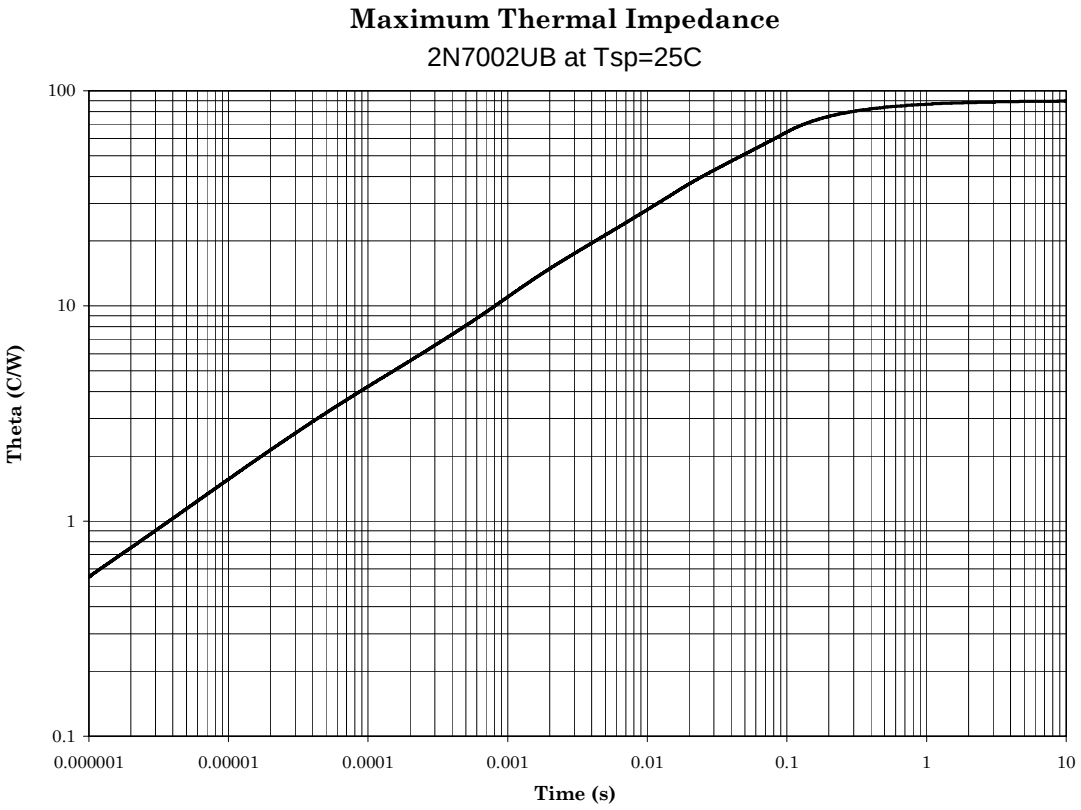


FIGURE 10. Thermal impedance graph ($R_{\theta JSP(IS)}$) for 2N7002UB (UB).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system Commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. The lead finish as specified (see 3.4.1).

d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer's List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43218-3990 or email vqe.chief@dla.mil .

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5961- 2742)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:52Z
Check the source to verify that this is the current version before use.